CURRICULUM VITAE

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Research Background: Link

Biography

Vatsanai Jaiton (Oh) was born in Yasothon, Thailand, in 1989. Currently, he is a Ph.D. candidate at the Bio-inspired Robotics and Neural Engineering laboratory (BRAIN Lab), School of Information Science and Technology (IST), Vidyasirimedhi Institute of Science and Technology (VISTEC), Thailand. He received his B. Eng. degree in instrumentation engineering with second-class honors from the King Mongkut's Institute of Technology Ladkrabang, Thailand, in 2015.

Research Interests

Unmanned Aerial Vehicles (UAVs); Autonomous Robots; Service and Inspection Robots; Neural Control for Navigation and Exploration; Neural Control for Aerial Robotics; Embodied AI and Neurorobotics; Embedded AI and Energy-Efficient Neural Computing Systems; Neural Dynamics, Learning, and Plasticity; Machine Learning for Robotics; Machine Learning and AI for Process Control and Automation; Intelligent Instrumentation and Control Systems.

Academic Background

Oct 2025 – May 2026
Visiting Ph.D. (7 months)
Research Center for Cognitive Computing in Life Sciences
Institute of Computational Life Sciences (ICLS)
Zurich University of Applied Sciences (ZHAW), Wädenswil, Switzerland

2018 – 2026
Ph.D. in Information Science and Technology (Defense completed on 30 January 2026)
Bio-inspired Robotics and Neural Engineering Laboratory (BRAIN Lab),
School of Information Science and Technology (IST),
Vidyasirimedhi Institute of Science and Technology (VISTEC), Rayong, Thailand

2011 – 2015
B.Eng. degree in instrumentation engineering with second-class honors
King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand

2002 – 2008
High School at Mahachanachai Wittayakom school, Yasothon, Thailand

Work Experiences

August 2015 - June 2018

Instrument and Control Systems Engineer at SCG Chemicals

- Maintained and enhanced instrumentation and control system performance in a petrochemical plant to improve operational reliability and process stability.
- Planned and managed routine and turnaround maintenance activities, including technical support, troubleshooting, spare parts management, manpower allocation, cost control, and documentation.
- Managed multiple engineering and improvement projects, including Continuous Emissions Monitoring System (CEMS) revamp, ESD voting system modification, and Safety PLC upgrades.
- Led shutdown and turnaround execution teams, coordinating contractors, spare parts, calibration of transmitters, and overhaul of control valves and process analyzers.
- Implemented system modifications and reliability improvement initiatives to enhance plant safety, compliance, and operational efficiency.

2008 – 2011

Lab Operator in Refrigeration Systems Calorimeter Laboratory at Siam Compressor Industry (SCI)

- Conducted calorimeter testing and performance evaluation of refrigeration systems (air conditioners, chillers, and heat pumps) in a laboratory environment.
- Awarded Kobetsu Kaizen (2007) for the project "Energy Saving in Process Testing," which reduced laboratory electrical consumption by over 20% annually and established a scalable model adaptable to other laboratories.

List of Publications

T. Phetpoon*, V. Jaiton*, K. Rothomphiwat*, M. Manawakul*, P. Chirathanyanon, P. Ritmetee, P. Manoonpong, "**Multimodal Obstacle Detection and Adaptive Neural Control for Autonomous Drones**," *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Hangzhou, China, 2025, pp. 3495-3502, doi: 10.1109/IROS60139.2025.11246601. (* Equal contribution authors)

Jaiton, V., Jadhav, P. D., Rothomphiwat, K., Phetpoon, T., Tarapongniwat, K., Manoonpong, P. (2025) **An Integrated Adaptive System for On-Pipe Landing and Adherence of Autonomous Inspection Drones**, IEEE/SICE International Symposium on System Integration (SII) 2025, https://ieeexplore.ieee.org/document/10871060.

Jaiton, V., Manoonpong, P. (2025). **Neural Chaotic Dynamics for Adaptive Exploration Control of an Autonomous Flying Robot.** In: From Animals to Animats 17. SAB 2024. Lecture Notes in Computer Science, vol 14993. Springer, Cham. https://doi.org/10.1007/978-3-031-71533-4_19.

Jaiton, V., Rothomphiwat, K., Phetpoon, T., Manawakul, M., Manoonpong, P. (2024) **An Integrated Adaptive Control System for Obstacle Detection and Online Speed Adaptation of Autonomous Drones**, IEEE/SICE International Symposium on System Integration (SII) 2024, Ha Long, Vietnam, 2024, pp. 1355-1356, DOI: 10.1109/SII58957.2024.10417272.

Mingchinda, N., Jaiton, V., Leung, B., Manoonpong, P. (2023) **Leg-body coordination strategies for obstacle avoidance and narrow space navigation of multi-segmented, legged robots**, Frontiers in Neurorobotics, DOI: 10.3389/fnbot.2023.1214248 (link) (JIF=3.1, SCI, Q2)

Jaiton, V., Manoonpong, P. (2023) **Neuro-Lorenz Oscillator with Bias Adaptation for Adaptive Searching and Exploring Behaviors of Flying Systems**. Proceedings of The 11th International Symposium on Adaptive Motion of Animals and Machines (AMAM2023). 2023, p. 113-114, DOI: https://doi.org/10.18910/92293

Mingchinda, N.; Jaiton, V.; Leung, B.; and Manoonpong, P. (2022) **Neural Body Bending Control with Temporal Delays for Millipede-Like Turning Behaviour of a Multi-Segmented, Legged Robot,** From Animals to Animats 16. SAB 2022. Lecture Notes in Computer Science, pp. 52-63 (EI), DOI: https://link.springer.com/chapter/10.1007/978-3-031-16770-6_5

Jaiton, V., Rothomphiwat, K., Ebeid E.S., and Manoonpong, P. (2022) **Neural Control and Online Learning for Speed Adaptation of Unmanned Aerial Vehicles**, Frontiers in Neural Circuits, DOI: https://doi.org/10.3389/fncir.2022.839361 (JIF = 3.492, SJR = 1.91, SCIE, Q1)

Jaiton, V., Manoonpong, P. (2021) **Chaotic Neural Oscillator for Navigation and Exploration of Autonomous Drones,** Proceedings of the 9.5th International Symposium on Adaptive Motion of Animals and Machines (AMAM2021), DOI: https://doi.org/10.18910/84868

Honors & Awards

- Full Ph.D. Scholarship from Vidyasirimedhi Institute of Science and Technology (VISTEC)
- Best Senior Project Award (2014)
- Chevron Internship Scholarship: on an appraisal of working performance (2014)
- DOW Chemicals Internship Scholarship (2013 - 2014)
- Excellent Scholarship: KMITL (2013)
- Chumbhot - Pantip Foundation Scholarship (2012 - 2014)
- KMITL Scholarship (2011 -2014)
- Toshiba Foundation Scholarship (2011 - 2014)